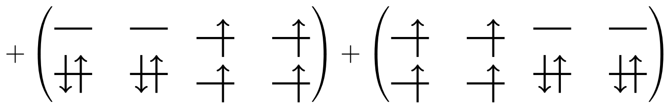
1 Linear Combinations

$$S * Q + Q * S$$



written in CI vectors:
monomer ci vectors:

$$+1 \cdot \ |0220| \cdot \ |aaaa| + 1 \cdot \ |aaaa| \cdot \ |0220|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+ \quad 1 \quad \cdot \quad \left|\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| \quad + \quad 1 \quad \cdot \quad \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\right|$$

substitution by dimer:

$$+1 \cdot \ |\left(-a_g + b_{1u}\right)\left(-b_{3u} + b_{2g}\right)\left(-b_{2u} + b_{3g}\right)\left(-b_{1g} + a_u\right)| + 1 \cdot \ |\left(+a_g + b_{1u}\right)\left(+b_{3u} + b_{2g}\right)\left(+b_{2u} + b_{3g}\right)\left(+b_{1g} + a_u\right)|$$

multiplied out:

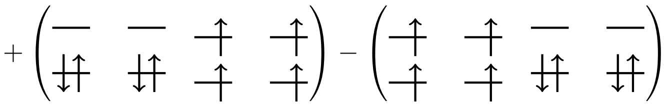
$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$
$$+\left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$
$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right|$$
$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$
$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right|$$
$$+\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

summarized:

$$+2 \cdot \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + 2 \cdot \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + 2 \cdot \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + 2 \cdot \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + 2 \cdot \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + 2 \cdot \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right|$$

$$+2 \cdot \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + 2 \cdot \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$S * Q - Q * S$$



written in CI vectors:
monomer ci vectors:

$$+1 \cdot |0220| \cdot |aaaa| - 1 \cdot |aaaa| \cdot |0220|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+ \quad 1 \quad \cdot \quad \left|\left(b_{1u}^r\right)\left(b_{2g}^r\right)\left(b_{3g}^r\right)\left(a_u^r\right)\right| \quad - \quad 1 \quad \cdot \quad \left|\left(b_{1u}^l\right)\left(b_{2g}^l\right)\left(b_{3g}^l\right)\left(a_u^l\right)\right|$$

substitution by dimer:

$$+1 \cdot |(-a_g + b_{1u})(-b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(+b_{2u} + b_{3g})(+b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$
$$+\left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$
$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right|$$
$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$
$$-\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right|$$
$$-\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

summarized:

$$-2\cdot\left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - 2\cdot\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - 2\cdot\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - 2\cdot\left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$
$$-2\cdot\left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$i^3 b_{2u} * i^3 b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:
monomer ci vectors:

$$+1\cdot\,|a2a0|\cdot\,|a2a0| - 1\cdot\,|a2a0|\cdot\,|0a2a| - 1\cdot\,|0a2a|\cdot\,|a2a0| + 1\cdot\,|0a2a|\cdot\,|0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1\cdot\left|\left(b^{l}_{1u}\right)\left(b^{l}_{3g}\right)\left(b^{r}_{1u}\right)\left(b^{r}_{3g}\right)\right|-1\cdot\left|\left(b^{l}_{1u}\right)\left(b^{l}_{3g}\right)\left(b^{r}_{2g}\right)\left(a^{r}_{u}\right)\right|-1\cdot\left|\left(b^{l}_{2g}\right)\left(a^{l}_{u}\right)\left(b^{r}_{1u}\right)\left(b^{r}_{3g}\right)\right|+1\cdot\left|\left(b^{l}_{2g}\right)\left(a^{l}_{u}\right)\left(b^{r}_{2g}\right)\left(a^{r}_{u}\right)\right|$$

substitution by dimer:

$$+1\cdot\left|\left(+a_{g}+b_{1u}\right)\left(+b_{2u}+b_{3g}\right)\left(-a_{g}+b_{1u}\right)\left(-b_{2u}+b_{3g}\right)\right|-1\cdot\left|\left(+a_{g}+b_{1u}\right)\left(+b_{2u}+b_{3g}\right)\left(-b_{3u}+b_{2g}\right)\left(-b_{1g}+a_{u}\right)\right|$$

$$-1\cdot\left|\left(+b_{3u}+b_{2g}\right)\left(+b_{1g}+a_{u}\right)\left(-a_{g}+b_{1u}\right)\left(-b_{2u}+b_{3g}\right)\right|+1\cdot\left|\left(+b_{3u}+b_{2g}\right)\left(+b_{1g}+a_{u}\right)\left(-b_{3u}+b_{2g}\right)\left(-b_{1g}+a_{u}\right)\right|$$

multiplied out:

$$
\begin{aligned}
&+\Big|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\Big| - \Big|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\Big| - \Big|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\Big| + \Big|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\Big| + \Big|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\Big| - \Big|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\Big| \\
&-\Big|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\Big| + \Big|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\Big| + \Big|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\Big| - \Big|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\Big| - \Big|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\Big| + \Big|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\Big| \\
&+\Big|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\Big| - \Big|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\Big| - \Big|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\Big| + \Big|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\Big| - \Big|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\Big| + \Big|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\Big| \\
&+\Big|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\Big| - \Big|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\Big| - \Big|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\Big| + \Big|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\Big| + \Big|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\Big| - \Big|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\Big| \\
&-\Big|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\Big| + \Big|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\Big| + \Big|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\Big| - \Big|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\Big| - \Big|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\Big| + \Big|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\Big| \\
&+\Big|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\Big| - \Big|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\Big| - \Big|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\Big| + \Big|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\Big| + \Big|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\Big| - \Big|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\Big| \\
&-\Big|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\Big| + \Big|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\Big| + \Big|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\Big| - \Big|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\Big| - \Big|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\Big| + \Big|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\Big| \\
&+\Big|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\Big| - \Big|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\Big| - \Big|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\Big| + \Big|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\Big| + \Big|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\Big| - \Big|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\Big| \\
&+\Big|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\Big| - \Big|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\Big| - \Big|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\Big| + \Big|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\Big| + \Big|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\Big| - \Big|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\Big| \\
&-\Big|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\Big| + \Big|\underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}}\Big| + \Big|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\Big| - \Big|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\Big| - \Big|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\Big| + \Big|\underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}}\Big| \\
&+\Big|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\Big| - \Big|\underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}}\Big| - \Big|\underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}}\Big| + \Big|\underbrace{b_{2g} b_{2g} a_u a_u}_{a_g}\Big|
\end{aligned}
$$

summarized:

$$
\begin{aligned}
&+\Big|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\Big| - \Big|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\Big| - \Big|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\Big| + \Big|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\Big| - 2\cdot\Big|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\Big| - 2\cdot\Big|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\Big| \\
&+2\cdot\Big|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\Big| + 2\cdot\Big|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\Big| + 2\cdot\Big|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\Big| + 2\cdot\Big|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\Big| - 2\cdot\Big|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\Big| - 2\cdot\Big|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\Big| \\
&+\Big|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\Big| - \Big|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\Big| - \Big|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\Big| + \Big|\underbrace{b_{2g} b_{2g} a_u a_u}_{a_g}\Big|
\end{aligned}
$$

$$
i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u}
$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:
monomer ci vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a|$$

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tbitbitbitbitbitbi

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tbitbitbitbitbitbi

tbitbitbitbitbitbi

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right|$$

$$-1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})| + 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})| - 1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}}\right|$$

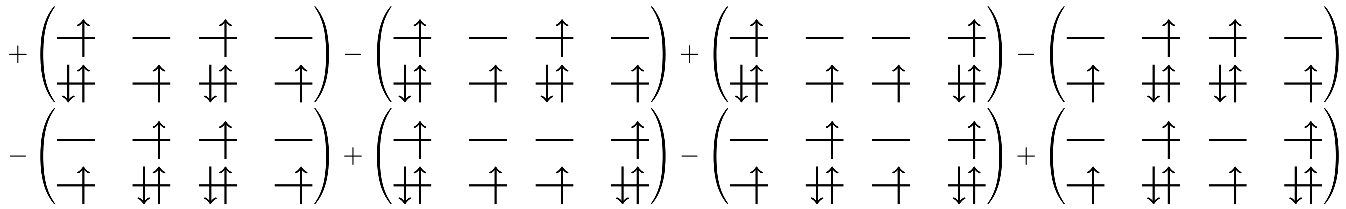
$$-\left|b_{3u} b_{1g} b_{2g} a_u\right| + \left|b_{3u} b_{2g} a_u a_u\right| + \left|b_{1g} b_{2g} b_{2g} a_u\right| - \left|b_{2g} b_{2g} a_u a_u\right| - \left|b_{3u} b_{3u} b_{1g} b_{1g}\right| + \left|b_{3u} b_{3u} b_{1g} a_u\right|$$

summarized:

$$+2 \cdot \left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| - 2 \cdot \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right| - 2 \cdot \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| + 2 \cdot \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| - 2 \cdot \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\right| + 2 \cdot \left|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\right|$$

$$+2 \cdot \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\right| - 2 \cdot \left|\underbrace{b_{2g} b_{2g} a_u a_u}_{a_g}\right|$$

$$i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |a2a0|$$

$$-1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$-1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1\cdot|(+a_g+b_{1u})(+b_{2u}+b_{3g})(-a_g+b_{1u})(-b_{2u}+b_{3g})| - 1\cdot|(+a_g+b_{1u})(+b_{2u}+b_{3g})(-a_g+b_{1u})(-b_{2u}+b_{3g})|$$

$$+1\cdot|(+a_g+b_{1u})(+b_{2u}+b_{3g})(-b_{3u}+b_{2g})(-b_{1g}+a_u)| - 1\cdot|(+b_{3u}+b_{2g})(+b_{1g}+a_u)(-a_g+b_{1u})(-b_{2u}+b_{3g})|$$

$$-1\cdot|(+b_{3u}+b_{2g})(+b_{1g}+a_u)(-a_g+b_{1u})(-b_{2u}+b_{3g})| + 1\cdot|(+a_g+b_{1u})(+b_{2u}+b_{3g})(-b_{3u}+b_{2g})(-b_{1g}+a_u)|$$

$$-1\cdot|(+b_{3u}+b_{2g})(+b_{1g}+a_u)(-b_{3u}+b_{2g})(-b_{1g}+a_u)| + 1\cdot|(+b_{3u}+b_{2g})(+b_{1g}+a_u)(-b_{3u}+b_{2g})(-b_{1g}+a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}}\right|$$

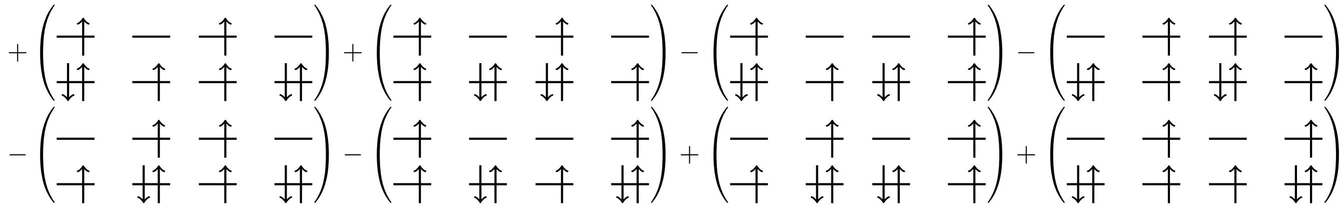
$$-\left|b_{3u} b_{1g} b_{2g} a_u\right| + \left|b_{3u} b_{2g} a_u a_u\right| + \left|b_{1g} b_{2g} b_{2g} a_u\right| - \left|b_{2g} b_{2g} a_u a_u\right| + \left|b_{3u} b_{3u} b_{1g} b_{1g}\right| - \left|b_{3u} b_{3u} b_{1g} a_u\right|$$

summarized:

$$-4\cdot\left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - 4\cdot\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + 4\cdot\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + 4\cdot\left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + 4\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + 4\cdot\left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right|$$

$$-4\cdot\left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - 4\cdot\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1\cdot\,|a2a0|\cdot\,|aa20| + 1\cdot\,|aa20|\cdot\,|a2a0| - 1\cdot\,|a2a0|\cdot\,|02aa| - 1\cdot\,|02aa|\cdot\,|a2a0|$$

$$-1\cdot\,|0a2a|\cdot\,|aa20| - 1\cdot\,|aa20|\cdot\,|0a2a| + 1\cdot\,|0a2a|\cdot\,|02aa| + 1\cdot\,|02aa|\cdot\,|0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right|$$

$$-1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

substitution by dimer:

$$+1\cdot|(+a_g+b_{1u})(+b_{2u}+b_{3g})(-a_g+b_{1u})(-b_{3u}+b_{2g})|+1\cdot|(+a_g+b_{1u})(+b_{3u}+b_{2g})(-a_g+b_{1u})(-b_{2u}+b_{3g})|$$

$$-1\cdot|(+a_g+b_{1u})(+b_{2u}+b_{3g})(-b_{2u}+b_{3g})(-b_{1g}+a_u)|-1\cdot|(+b_{2u}+b_{3g})(+b_{1g}+a_u)(-a_g+b_{1u})(-b_{2u}+b_{3g})|$$

$$-1\cdot|(+b_{3u}+b_{2g})(+b_{1g}+a_u)(-a_g+b_{1u})(-b_{3u}+b_{2g})|-1\cdot|(+a_g+b_{1u})(+b_{3u}+b_{2g})(-b_{3u}+b_{2g})(-b_{1g}+a_u)|$$

$$+1\cdot|(+b_{3u}+b_{2g})(+b_{1g}+a_u)(-b_{2u}+b_{3g})(-b_{1g}+a_u)|+1\cdot|(+b_{2u}+b_{3g})(+b_{1g}+a_u)(-b_{3u}+b_{2g})(-b_{1g}+a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

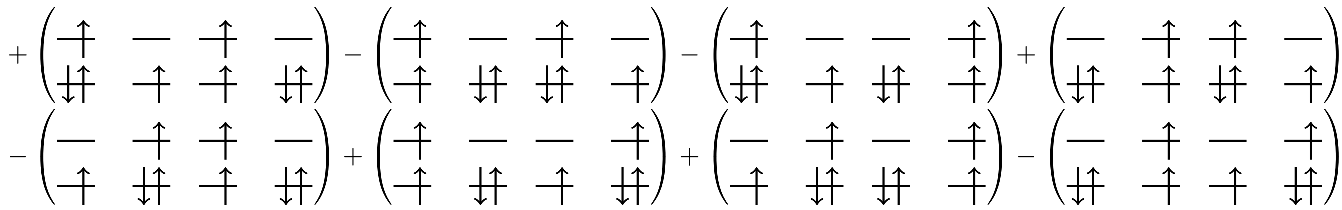
$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

summarized:

$$
\begin{aligned}
&+2\cdot\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| \\
&+2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| \\
&+2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}}\right|
\end{aligned}
$$

$$i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u}$$



written in CI vectors:

monomer ci vectors:

$$+1\cdot|a2a0|\cdot|aa20| - 1\cdot|aa20|\cdot|a2a0| - 1\cdot|a2a0|\cdot|02aa| + 1\cdot|02aa|\cdot|a2a0|$$

$$-1\cdot|0a2a|\cdot|aa20| + 1\cdot|aa20|\cdot|0a2a| + 1\cdot|0a2a|\cdot|02aa| - 1\cdot|02aa|\cdot|0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$-1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})\,(+b_{2u} + b_{3g})\,(-a_g + b_{1u})\,(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})\,(+b_{3u} + b_{2g})\,(-a_g + b_{1u})\,(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})\,(+b_{2u} + b_{3g})\,(-b_{2u} + b_{3g})\,(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})\,(+b_{1g} + a_u)\,(-a_g + b_{1u})\,(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})\,(+b_{1g} + a_u)\,(-a_g + b_{1u})\,(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})\,(+b_{3u} + b_{2g})\,(-b_{3u} + b_{2g})\,(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})\,(+b_{1g} + a_u)\,(-b_{2u} + b_{3g})\,(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})\,(+b_{1g} + a_u)\,(-b_{3u} + b_{2g})\,(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

summarized:

$$-2\cdot\left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - 2\cdot\left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| + 2\cdot\left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{2u} b_{2g} a_u a_u}_{a_u}\right|$$

$$i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u}$$

$$+\begin{pmatrix} \uparrow & \text{—} & \uparrow & \text{—} \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{—} & \uparrow & \text{—} \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & \text{—} & \text{—} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \text{—} & \uparrow & \uparrow & \text{—} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$-\begin{pmatrix} \text{—} & \uparrow & \uparrow & \text{—} \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & \text{—} & \text{—} & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \text{—} & \uparrow & \text{—} & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \text{—} & \uparrow & \text{—} & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI vectors:
monomer ci vectors:

$$+1\cdot|a2a0|\cdot|aa20| + 1\cdot|aa20|\cdot|a2a0| + 1\cdot|a2a0|\cdot|02aa| + 1\cdot|02aa|\cdot|a2a0|$$

$$-1\cdot|0a2a|\cdot|aa20| - 1\cdot|aa20|\cdot|0a2a| - 1\cdot|0a2a|\cdot|02aa| - 1\cdot|02aa|\cdot|0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$-1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

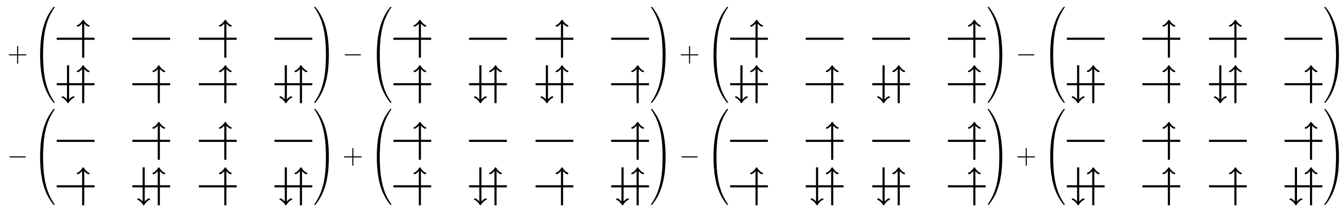
summarized:

$$+2\cdot\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}}\right|$$

$$i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1\cdot|a2a0|\cdot|aa20| - 1\cdot|aa20|\cdot|a2a0| + 1\cdot|a2a0|\cdot|02aa| - 1\cdot|02aa|\cdot|a2a0|$$

$$-1\cdot|0a2a|\cdot|aa20| + 1\cdot|aa20|\cdot|0a2a| - 1\cdot|0a2a|\cdot|02aa| + 1\cdot|02aa|\cdot|0a2a|$$

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tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$-1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1\cdot|(+a_g+b_{1u})\,(+b_{2u}+b_{3g})\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})|-1\cdot|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-a_g+b_{1u})\,(-b_{2u}+b_{3g})|$$

$$+1\cdot|(+a_g+b_{1u})\,(+b_{2u}+b_{3g})\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)|-1\cdot|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-a_g+b_{1u})\,(-b_{2u}+b_{3g})|$$

$$-1\cdot|(+b_{3u}+b_{2g})\,(+b_{1g}+a_u)\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})|+1\cdot|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-b_{3u}+b_{2g})\,(-b_{1g}+a_u)|$$

$$-1\cdot|(+b_{3u}+b_{2g})\,(+b_{1g}+a_u)\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)|+1\cdot|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-b_{3u}+b_{2g})\,(-b_{1g}+a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

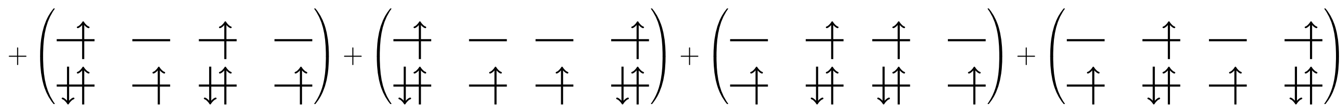
$$-\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

summarized:

$$-2 \cdot \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + 2 \cdot \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| + 2 \cdot \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - 2 \cdot \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| - 2 \cdot \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + 2 \cdot \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right|$$
$$+2 \cdot \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - 2 \cdot \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| + 2 \cdot \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - 2 \cdot \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - 2 \cdot \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + 2 \cdot \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$
$$+2 \cdot \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| - 2 \cdot \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| - 2 \cdot \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| + 2 \cdot \left|\underbrace{b_{2u} b_{2g} a_u a_u}_{a_u}\right|$$

$$e^3 b_{2u} * e^3 b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

substitution by dimer:

$$+1\cdot\left|(+a_g+b_{1u})\,(+b_{2u}+b_{3g})\,(-a_g+b_{1u})\,(-b_{2u}+b_{3g})\right| + 1\cdot\left|(+a_g+b_{1u})\,(+b_{2u}+b_{3g})\,(-b_{3u}+b_{2g})\,(-b_{1g}+a_u)\right|$$

$$+1\cdot\left|(+b_{3u}+b_{2g})\,(+b_{1g}+a_u)\,(-a_g+b_{1u})\,(-b_{2u}+b_{3g})\right| + 1\cdot\left|(+b_{3u}+b_{2g})\,(+b_{1g}+a_u)\,(-b_{3u}+b_{2g})\,(-b_{1g}+a_u)\right|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{2u} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{3g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{2g} a_u a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{2g} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2g} b_{2g} a_u a_u}_{a_g}\right|$$

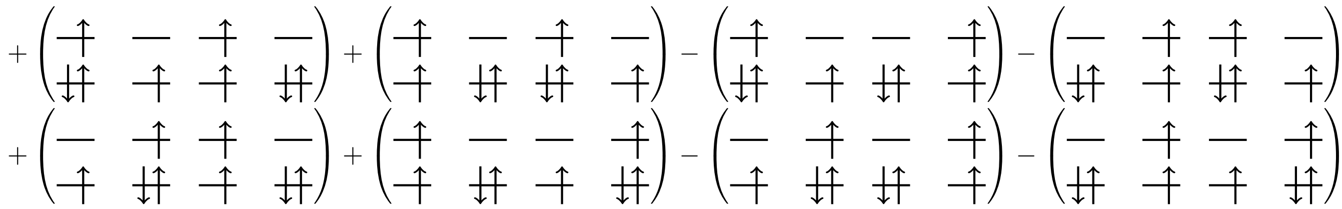
summarized:

$$+\left|\underbrace{a_g a_g b_{2u} b_{2u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3g} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{1u} b_{1u} b_{3g} b_{3g}}_{a_g}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + 2\cdot\left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - 2\cdot\left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - 2\cdot\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} a_u a_u}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2g} b_{2g} a_u a_u}_{a_g}\right|$$

$$e^3b_{2u} * e^3b_{3u} + e^3b_{3u} * e^3b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |02aa| \cdot |a2a0|$$

$$+1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |02aa| - 1 \cdot |02aa| \cdot |0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$+1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})\,(+b_{2u} + b_{3g})\,(-a_g + b_{1u})\,(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})\,(+b_{3u} + b_{2g})\,(-a_g + b_{1u})\,(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})\,(+b_{2u} + b_{3g})\,(-b_{2u} + b_{3g})\,(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})\,(+b_{1g} + a_u)\,(-a_g + b_{1u})\,(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})\,(+b_{1g} + a_u)\,(-a_g + b_{1u})\,(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})\,(+b_{3u} + b_{2g})\,(-b_{3u} + b_{2g})\,(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})\,(+b_{1g} + a_u)\,(-b_{2u} + b_{3g})\,(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})\,(+b_{1g} + a_u)\,(-b_{3u} + b_{2g})\,(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

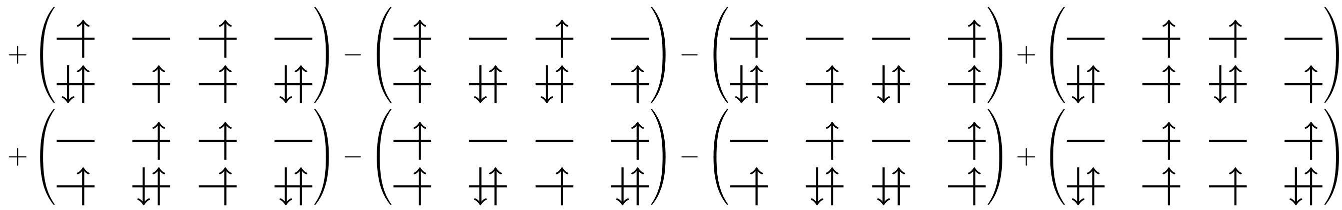
summarized:

$$+2\cdot\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right|$$

$$+2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}}\right|$$

$$e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1\cdot\ |a2a0|\cdot\ |aa20| - 1\cdot\ |aa20|\cdot\ |a2a0| - 1\cdot\ |a2a0|\cdot\ |02aa| + 1\cdot\ |02aa|\cdot\ |a2a0|$$

$$+1\cdot\ |0a2a|\cdot\ |aa20| - 1\cdot\ |aa20|\cdot\ |0a2a| - 1\cdot\ |0a2a|\cdot\ |02aa| + 1\cdot\ |02aa|\cdot\ |0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$+1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})\,(+b_{2u} + b_{3g})\,(-a_g + b_{1u})\,(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})\,(+b_{3u} + b_{2g})\,(-a_g + b_{1u})\,(-b_{2u} + b_{3g})|$$

$$-1 \cdot |(+a_g + b_{1u})\,(+b_{2u} + b_{3g})\,(-b_{2u} + b_{3g})\,(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})\,(+b_{1g} + a_u)\,(-a_g + b_{1u})\,(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})\,(+b_{1g} + a_u)\,(-a_g + b_{1u})\,(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})\,(+b_{3u} + b_{2g})\,(-b_{3u} + b_{2g})\,(-b_{1g} + a_u)|$$

$$-1 \cdot |(+b_{3u} + b_{2g})\,(+b_{1g} + a_u)\,(-b_{2u} + b_{3g})\,(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})\,(+b_{1g} + a_u)\,(-b_{3u} + b_{2g})\,(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

summarized:

$$
\begin{aligned}
&-2\cdot\left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - 2\cdot\left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| \\
&-2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| \\
&+2\cdot\left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| - 2\cdot\left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{2u} b_{2g} a_u a_u}_{a_u}\right|
\end{aligned}
$$

$$e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u}$$

$$
\begin{aligned}
&+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
&+\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}
$$

written in CI vectors:
monomer ci vectors:

$$+1\cdot|a2a0|\cdot|aa20| + 1\cdot|aa20|\cdot|a2a0| + 1\cdot|a2a0|\cdot|02aa| + 1\cdot|02aa|\cdot|a2a0|$$

$$+1\cdot|0a2a|\cdot|aa20| + 1\cdot|aa20|\cdot|0a2a| + 1\cdot|0a2a|\cdot|02aa| + 1\cdot|02aa|\cdot|0a2a|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{3g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{3g}^{r}\right)\right|$$

$$+1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{2g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{2g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| + 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

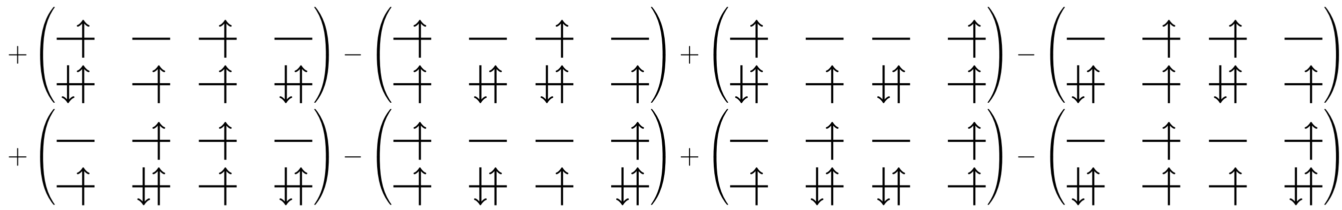
$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

summarized:

$$+2\cdot\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right|$$
$$-2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$
$$+2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right| - 2\cdot\left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| + 2\cdot\left|\underbrace{b_{2g} b_{3g} a_u a_u}_{b_{1g}}\right|$$

$$e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u}$$



written in CI vectors:
monomer ci vectors:

$$+1\cdot\ |a2a0|\cdot\ |aa20| - 1\cdot\ |aa20|\cdot\ |a2a0| + 1\cdot\ |a2a0|\cdot\ |02aa| - 1\cdot\ |02aa|\cdot\ |a2a0|$$

$$+1\cdot\ |0a2a|\cdot\ |aa20| - 1\cdot\ |aa20|\cdot\ |0a2a| + 1\cdot\ |0a2a|\cdot\ |02aa| - 1\cdot\ |02aa|\cdot\ |0a2a|$$

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tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

tbitbitbitbitbitbi

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right| + 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{3g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{3g}\right)\right|$$

$$+1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| - 1 \cdot \left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right| + 1 \cdot \left|\left(b^l_{2g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1 \cdot \left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{2g}\right)\left(a^r_{u}\right)\right|$$

substitution by dimer:

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+a_g + b_{1u})(+b_{2u} + b_{3g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{2u} + b_{3g})|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| - 1 \cdot |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

$$+1 \cdot |(+b_{3u} + b_{2g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)| - 1 \cdot |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{3u} + b_{2g})(-b_{1g} + a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2u}}_{b_{1g}}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g a_g b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} b_{1u}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u}\right|$$

$$+\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{2u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{2u} b_{2u} b_{1u} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{a_g b_{1g} b_{3g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{3g}}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u}\right| - \left|\underbrace{b_{1u} b_{3g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$

$$-\left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1g}}_{b_{1g}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u}\right| + \left|\underbrace{a_g b_{3u} b_{2g} a_u}_{b_{1g}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} a_u}_{b_{1g}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{2g}}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{1u} b_{2g} b_{2g} a_u}_{b_{1g}}\right|$$

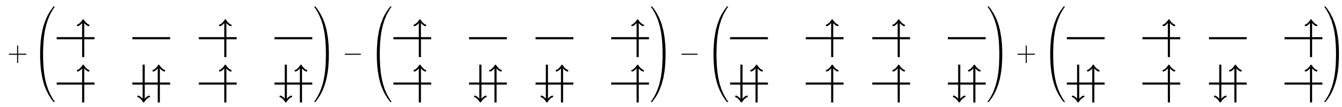
$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1g}}_{b_{1g}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| + \left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u}\right| - \left|\underbrace{b_{3u} b_{2u} a_u a_u}_{b_{1g}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{3g} a_u}_{b_{1g}}\right| + \left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| - \left|\underbrace{b_{2u} b_{1g} b_{2g} a_u}_{b_{1g}}\right| - \left|\underbrace{b_{1g} b_{1g} b_{2g} b_{3g}}_{b_{1g}}\right| + \left|\underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u}\right|$$

summarized:

$$-2\cdot\left|\underbrace{a_g a_g b_{2u} b_{2g}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g a_g b_{3u} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{2u} b_{1u} b_{1u} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{3u} b_{1u} b_{1u} b_{3g}}_{a_u}\right| - 2\cdot\left|\underbrace{a_g b_{2u} b_{2u} a_u}_{a_u}\right| + 2\cdot\left|\underbrace{a_g b_{3g} b_{3g} a_u}_{a_u}\right|$$
$$+2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1u}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{3g} b_{3g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1g} b_{1u}}_{a_u}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{3u} a_u}_{a_u}\right| + 2\cdot\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{a_g b_{2g} b_{2g} a_u}_{a_u}\right|$$
$$-2\cdot\left|\underbrace{b_{3u} b_{1g} b_{1g} b_{3g}}_{a_u}\right| + 2\cdot\left|\underbrace{b_{3u} b_{3g} a_u a_u}_{a_u}\right| + 2\cdot\left|\underbrace{b_{2u} b_{1g} b_{1g} b_{2g}}_{a_u}\right| - 2\cdot\left|\underbrace{b_{2u} b_{2g} a_u a_u}_{a_u}\right|$$

$$e^3 b_{3u} * e^3 b_{3u}$$



written in CI vectors:
monomer ci vectors:

$$+1\cdot\ |aa20|\cdot\ |aa20| - 1\cdot\ |aa20|\cdot\ |02aa| - 1\cdot\ |02aa|\cdot\ |aa20| + 1\cdot\ |02aa|\cdot\ |02aa|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1\cdot\left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| - 1\cdot\left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| - 1\cdot\left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1\cdot\left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right|$$

substitution by dimer:

$$+1\cdot\left|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})\right| - 1\cdot\left|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)\right|$$

$$-1\cdot\left|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})\right| + 1\cdot\left|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)\right|$$

multiplied out:

$$+\left|\underbrace{a_ga_gb_{3u}b_{3u}}_{a_g}\right| - \left|\underbrace{a_ga_gb_{3u}b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_gb_{3u}b_{3u}b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_gb_{3u}b_{1u}b_{2g}}_{a_g}\right| + \left|\underbrace{a_ga_gb_{3u}b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_ga_gb_{2g}b_{2g}}_{a_g}\right|$$

$$-\left|\underbrace{a_gb_{3u}b_{1u}b_{2g}}_{a_g}\right| + \left|\underbrace{a_gb_{1u}b_{2g}b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_gb_{3u}b_{3u}b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_gb_{3u}b_{1u}b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u}b_{3u}b_{1u}b_{1u}}_{a_g}\right| + \left|\underbrace{b_{3u}b_{1u}b_{1u}b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_gb_{3u}b_{1u}b_{2g}}_{a_g}\right| - \left|\underbrace{a_gb_{1u}b_{2g}b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u}b_{1u}b_{1u}b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u}b_{1u}b_{2g}b_{2g}}_{a_g}\right| - \left|\underbrace{a_gb_{3u}b_{2u}b_{1g}}_{a_g}\right| + \left|\underbrace{a_gb_{3u}b_{2u}a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_gb_{3u}b_{1g}b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_gb_{3u}b_{3g}a_u}_{a_g}\right| - \left|\underbrace{a_gb_{2u}b_{1g}b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_gb_{2u}b_{2g}a_u}_{a_g}\right| + \left|\underbrace{a_gb_{1g}b_{2g}b_{3g}}_{a_g}\right| - \left|\underbrace{a_gb_{2g}b_{3g}a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{b_{3u}b_{2u}b_{1g}b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u}b_{2u}b_{1u}a_u}_{a_g}\right| + \left|\underbrace{b_{3u}b_{1g}b_{1u}b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u}b_{1u}b_{3g}a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u}b_{1g}b_{1u}b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u}b_{1u}b_{2g}a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{1g}b_{1u}b_{2g}b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{1u}b_{2g}b_{3g}a_u}_{a_g}\right| - \left|\underbrace{a_gb_{3u}b_{2u}b_{1g}}_{a_g}\right| + \left|\underbrace{a_gb_{2u}b_{1g}b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{3u}b_{2u}b_{1g}b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{2u}b_{1g}b_{1u}b_{2g}}_{a_g}\right|$$

$$-\left|\underbrace{a_gb_{3u}b_{2u}a_u}_{b_{1u}}\right| + \left|\underbrace{a_gb_{2u}b_{2g}a_u}_{a_g}\right| + \left|\underbrace{b_{3u}b_{2u}b_{1u}a_u}_{a_g}\right| - \left|\underbrace{b_{2u}b_{1u}b_{2g}a_u}_{b_{1u}}\right| - \left|\underbrace{a_gb_{3u}b_{1g}b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_gb_{1g}b_{2g}b_{3g}}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u}b_{1g}b_{1u}b_{3g}}_{a_g}\right| - \left|\underbrace{b_{1g}b_{1u}b_{2g}b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_gb_{3u}b_{3g}a_u}_{a_g}\right| + \left|\underbrace{a_gb_{2g}b_{3g}a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u}b_{1u}b_{3g}a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u}b_{2g}b_{3g}a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{2u}b_{2u}b_{1g}b_{1g}}_{a_g}\right| - \left|\underbrace{b_{2u}b_{2u}b_{1g}a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u}b_{1g}b_{1g}b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{2u}b_{1g}b_{3g}a_u}_{a_g}\right| + \left|\underbrace{b_{2u}b_{2u}b_{1g}a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u}b_{2u}a_ua_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u}b_{1g}b_{3g}a_u}_{a_g}\right| + \left|\underbrace{b_{2u}b_{3g}a_ua_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u}b_{1g}b_{1g}b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u}b_{1g}b_{3g}a_u}_{a_g}\right| - \left|\underbrace{b_{1g}b_{1g}b_{3g}b_{3g}}_{a_g}\right| + \left|\underbrace{b_{1g}b_{3g}b_{3g}a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{2u}b_{1g}b_{3g}a_u}_{a_g}\right| - \left|\underbrace{b_{2u}b_{3g}a_ua_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g}b_{3g}b_{3g}a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3g}b_{3g}a_ua_u}_{a_g}\right|$$

summarized:

$$+\left|\underbrace{a_ga_gb_{3u}b_{3u}}_{a_g}\right| - \left|\underbrace{a_ga_gb_{2g}b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u}b_{3u}b_{1u}b_{1u}}_{a_g}\right| + \left|\underbrace{b_{1u}b_{1u}b_{2g}b_{2g}}_{a_g}\right| - 2\cdot\left|\underbrace{a_gb_{3u}b_{2u}b_{1g}}_{a_g}\right| - 2\cdot\left|\underbrace{a_gb_{3u}b_{3g}a_u}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{a_gb_{2u}b_{2g}a_u}_{a_g}\right| + 2\cdot\left|\underbrace{a_gb_{1g}b_{2g}b_{3g}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{3u}b_{2u}b_{1u}a_u}_{a_g}\right| + 2\cdot\left|\underbrace{b_{3u}b_{1g}b_{1u}b_{3g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{2u}b_{1g}b_{1u}b_{2g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{1u}b_{2g}b_{3g}a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{2u}b_{2u}b_{1g}b_{1g}}_{a_g}\right| - \left|\underbrace{b_{2u}b_{2u}a_ua_u}_{a_g}\right| - \left|\underbrace{b_{1g}b_{1g}b_{3g}b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3g}b_{3g}a_ua_u}_{a_g}\right|$$

$$e^3b_{3u} * i^3b_{3u} + i^3b_{3u} * e^3b_{3u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

written in CI vectors:
monomer ci vectors:

$$+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |aa20|$$

$$-1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right|$$

$$-1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1\cdot\left|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})\right|+1\cdot\left|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})\right|$$

$$+1\cdot\left|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)\right|+1\cdot\left|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})\right|$$

$$-1\cdot\left|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-a_g+b_{1u})\,(-b_{3u}+b_{2g})\right|-1\cdot\left|(+a_g+b_{1u})\,(+b_{3u}+b_{2g})\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)\right|$$

$$-1\cdot\left|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)\right|-1\cdot\left|(+b_{2u}+b_{3g})\,(+b_{1g}+a_u)\,(-b_{2u}+b_{3g})\,(-b_{1g}+a_u)\right|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{2u} a_u a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{3g} a_u a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{3g} b_{3g} a_u}_{b_{1u}}\right|$$

$$-\left|b_{2u} b_{1g} b_{3g} a_u\right| + \left|b_{2u} b_{3g} a_u a_u\right| + \left|b_{1g} b_{3g} b_{3g} a_u\right| - \left|b_{3g} b_{3g} a_u a_u\right| - \left|b_{2u} b_{2u} b_{1g} b_{1g}\right| + \left|b_{2u} b_{2u} b_{1g} a_u\right|$$

summarized:

$$+2\cdot\left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| - 2\cdot\left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{2u} b_{2u} a_u a_u}_{a_g}\right|$$
$$+2\cdot\left|\underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{3g} b_{3g} a_u a_u}_{a_g}\right|$$

$$e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u}$$

$$+\begin{pmatrix}\uparrow & \text{---} & \uparrow & \text{---}\\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow\end{pmatrix} - \begin{pmatrix}\uparrow & \text{---} & \uparrow & \text{---}\\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow\end{pmatrix} + \begin{pmatrix}\uparrow & \text{---} & \text{---} & \uparrow\\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow\end{pmatrix} - \begin{pmatrix}\text{---} & \uparrow & \uparrow & \text{---}\\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow\end{pmatrix}$$
$$-\begin{pmatrix}\text{---} & \uparrow & \uparrow & \text{---}\\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow\end{pmatrix} + \begin{pmatrix}\uparrow & \text{---} & \text{---} & \uparrow\\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow\end{pmatrix} - \begin{pmatrix}\text{---} & \uparrow & \text{---} & \uparrow\\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow\end{pmatrix} + \begin{pmatrix}\text{---} & \uparrow & \text{---} & \uparrow\\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow\end{pmatrix}$$

written in CI vectors:
monomer ci vectors:

$$+1\cdot|aa20|\cdot|aa20| - 1\cdot|aa20|\cdot|aa20| + 1\cdot|aa20|\cdot|02aa| - 1\cdot|02aa|\cdot|aa20|$$

$$-1\cdot|02aa|\cdot|aa20| + 1\cdot|aa20|\cdot|02aa| - 1\cdot|02aa|\cdot|02aa| + 1\cdot|02aa|\cdot|02aa|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| - 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right|$$

$$-1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{1u}^{r}\right)\left(b_{2g}^{r}\right)\right| + 1 \cdot \left|\left(b_{1u}^{l}\right)\left(b_{2g}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| - 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right| + 1 \cdot \left|\left(b_{3g}^{l}\right)\left(a_{u}^{l}\right)\left(b_{3g}^{r}\right)\left(a_{u}^{r}\right)\right|$$

substitution by dimer:

$$+1\cdot|(+a_g+b_{1u})(+b_{3u}+b_{2g})(-a_g+b_{1u})(-b_{3u}+b_{2g})|-1\cdot|(+a_g+b_{1u})(+b_{3u}+b_{2g})(-a_g+b_{1u})(-b_{3u}+b_{2g})|$$

$$+1\cdot|(+a_g+b_{1u})(+b_{3u}+b_{2g})(-b_{2u}+b_{3g})(-b_{1g}+a_u)|-1\cdot|(+b_{2u}+b_{3g})(+b_{1g}+a_u)(-a_g+b_{1u})(-b_{3u}+b_{2g})|$$

$$-1\cdot|(+b_{2u}+b_{3g})(+b_{1g}+a_u)(-a_g+b_{1u})(-b_{3u}+b_{2g})|+1\cdot|(+a_g+b_{1u})(+b_{3u}+b_{2g})(-b_{2u}+b_{3g})(-b_{1g}+a_u)|$$

$$-1\cdot|(+b_{2u}+b_{3g})(+b_{1g}+a_u)(-b_{2u}+b_{3g})(-b_{1g}+a_u)|+1\cdot|(+b_{2u}+b_{3g})(+b_{1g}+a_u)(-b_{2u}+b_{3g})(-b_{1g}+a_u)|$$

multiplied out:

$$+\left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right|$$

$$-\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right|$$

$$+\left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right|$$

$$+\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right|$$

$$-\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{2u} a_u a_u}_{a_g}\right|$$

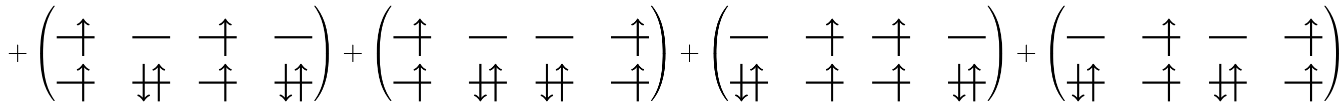
$$+\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{3g} a_u a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g}\right| - \left|\underbrace{b_{1g} b_{3g} b_{3g} a_u}_{b_{1u}}\right|$$

$$-\left|b_{2u} b_{1g} b_{3g} a_u\right| + \left|b_{2u} b_{3g} a_u a_u\right| + \left|b_{1g} b_{3g} b_{3g} a_u\right| - \left|b_{3g} b_{3g} a_u a_u\right| + \left|b_{2u} b_{2u} b_{1g} b_{1g}\right| - \left|b_{2u} b_{2u} b_{1g} a_u\right|$$

summarized:

$$-4\cdot\left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - 4\cdot\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + 4\cdot\left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| + 4\cdot\left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| + 4\cdot\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + 4\cdot\left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right|$$
$$-4\cdot\left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| - 4\cdot\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right|$$

$$i^3 b_{3u} * i^3 b_{3u}$$



written in CI vectors:
monomer ci vectors:

$$+1\cdot\ |aa20|\cdot\ |aa20| + 1\cdot\ |aa20|\cdot\ |02aa| + 1\cdot\ |02aa|\cdot\ |aa20| + 1\cdot\ |02aa|\cdot\ |02aa|$$

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or, expressed in terms of symmetry:
monomer determinants:

$$+1\cdot\left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1\cdot\left|\left(b^l_{1u}\right)\left(b^l_{2g}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right| + 1\cdot\left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{1u}\right)\left(b^r_{2g}\right)\right| + 1\cdot\left|\left(b^l_{3g}\right)\left(a^l_{u}\right)\left(b^r_{3g}\right)\left(a^r_{u}\right)\right|$$

substitution by dimer:

$$+1\cdot\ |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1\cdot\ |(+a_g + b_{1u})(+b_{3u} + b_{2g})(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

$$+1\cdot\ |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-a_g + b_{1u})(-b_{3u} + b_{2g})| + 1\cdot\ |(+b_{2u} + b_{3g})(+b_{1g} + a_u)(-b_{2u} + b_{3g})(-b_{1g} + a_u)|$$

multiplied out:

$$
\begin{aligned}
&+\left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g a_g b_{3u} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right| \\
&-\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3u} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| \\
&+\left|\underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{a_g b_{1u} b_{2g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{1u} b_{2g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| \\
&-\left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| + \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| \\
&+\left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| \\
&-\left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| - \left|\underbrace{a_g b_{2u} b_{1g} b_{2g}}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1g} b_{1u}}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| \\
&+\left|\underbrace{a_g b_{3u} b_{2u} a_u}_{b_{1u}}\right| - \left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - \left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| + \left|\underbrace{b_{2u} b_{1u} b_{2g} a_u}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| \\
&-\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{1g} b_{1u} b_{2g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{a_g b_{2g} b_{3g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{3u} b_{1u} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| \\
&+\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{b_{2u} b_{2u} b_{1g} a_u}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{2u} a_u a_u}_{a_g}\right| \\
&-\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| + \left|\underbrace{b_{2u} b_{3g} a_u a_u}_{b_{1u}}\right| + \left|\underbrace{b_{2u} b_{1g} b_{1g} b_{3g}}_{b_{1u}}\right| - \left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{1g} b_{3g} b_{3g} a_u}_{b_{1u}}\right| \\
&+\left|\underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g}\right| - \left|\underbrace{b_{2u} b_{3g} a_u a_u}_{b_{1u}}\right| - \left|\underbrace{b_{1g} b_{3g} b_{3g} a_u}_{b_{1u}}\right| + \left|\underbrace{b_{3g} b_{3g} a_u a_u}_{a_g}\right|
\end{aligned}
$$

summarized:

$$
\begin{aligned}
&+\left|\underbrace{a_g a_g b_{3u} b_{3u}}_{a_g}\right| - \left|\underbrace{a_g a_g b_{2g} b_{2g}}_{a_g}\right| - \left|\underbrace{b_{3u} b_{3u} b_{1u} b_{1u}}_{a_g}\right| + \left|\underbrace{b_{1u} b_{1u} b_{2g} b_{2g}}_{a_g}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g}\right| + 2\cdot\left|\underbrace{a_g b_{3u} b_{3g} a_u}_{a_g}\right| \\
&-2\cdot\left|\underbrace{a_g b_{2u} b_{2g} a_u}_{a_g}\right| - 2\cdot\left|\underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g}\right| - 2\cdot\left|\underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g}\right| - 2\cdot\left|\underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g}\right| + 2\cdot\left|\underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g}\right| \\
&+\left|\underbrace{b_{2u} b_{2u} b_{1g} b_{1g}}_{a_g}\right| - \left|\underbrace{b_{2u} b_{2u} a_u a_u}_{a_g}\right| - \left|\underbrace{b_{1g} b_{1g} b_{3g} b_{3g}}_{a_g}\right| + \left|\underbrace{b_{3g} b_{3g} a_u a_u}_{a_g}\right|
\end{aligned}
$$
